

EDUCATION

- 2024 **Doctor of Philosophy (Molecular, Cellular and Integrative Biosciences)**
The Pennsylvania State University, University Park, PA
• Concentration in functional, computational, and evolutionary genomics with an emphasis in gene-environment interactions and parent-of-origin effects
- 2019 **Bachelor of Science (Neuroscience)**
The University of Nebraska at Omaha, Omaha, NE

RESEARCH AND WORK EXPERIENCE

- March 2026 – **Associate Data Scientist**
Present; *The University of Texas MD Anderson Cancer Center, Houston, TX (Remote)*
Institute for Data Science in Oncology, Department of Epidemiology
Previously: *Assistant Data Scientist, July 2024 – March 2026*
- Applied isoform-resolution transcriptome-wide association study (TWAS) framework integrating multi-tissue GTEx RNA-seq with population-scale GWAS across cancer indications, integrating ENCODE datasets to identify molecular mechanisms associated with lead isoQTLs
 - Assembled tissue-specific transcriptome references using Oxford Nanopore and PacBio long-read RNA-seq, novel isoforms and reducing short-read quantification uncertainty; demonstrated improved eQTL detection and TWAS accuracy in colocalization studies
 - Conducted systematic comparative analyses of RNA-seq quantification methods and transcriptomic annotations affecting replicability of eQTL detection, TWAS, and colocalization results
 - Established multi-ancestry placental genomics resource across multiple birth cohorts to investigate gene-environment interactions underlying childhood metabolic disease, including genome-wide GxE scans testing how gestational diabetes and PFAS exposure modify fetal genetic regulation
 - Characterized parent-of-origin-specific (genomic imprinting) regulation at isoform level using trio genotypes, revealing allele-specific expression disruption by environmental exposures and demonstrating that natural variation in transplacental transfer efficiency exposes distinct transcriptional networks mediating PFAS effects on birth outcomes
 - Developed ICONIC R package for causal inference in observational multiomics, integrating genetic instruments and negative controls to diagnose, estimate, and stress-test mediation effects under unmeasured confounding using data-calibrated generative models
- March 2026 – **Bioinformatics Data Science Contractor**
Present *Vindhya Data Science (Remote)*
- Conducted pan-cancer genomic analyses across TCGA cohorts and DepMap CRISPR screens to evaluate therapeutic target candidates across multiple cancer indications
- June – August 2024 **Data, AI & Genome Sciences Intern**
Merck Research Laboratories, Boston, MA
- Conducted analyses differentiating hematopoietic and tumor mutations in liquid biopsies leveraging KEYTRUDA™ keynote study genomics data
- 2019 – 2024 **National Science Foundation Graduate Research Fellow**
2019 – 2022 **Integrative Pollinator Ecology Graduate Training Fellow**
2019 – 2020 **Penn State University Graduate Fellow**
The Pennsylvania State University, Huck Institutes of the Life Sciences, University Park, PA
- Generated multi-omics datasets and used statistical genomics and machine-learning methods to study molecular mechanisms of parental conflicts underlying plasticity in bee behaviors
 - Analyzed tissue-specific gene expression profiles to investigate phenotypic plasticity
 - Designed metagenomics software to assess species biodiversity in environmental samples

RESEARCH AND WORK EXPERIENCE (cont.)

- 2018 – 2019 **Research Assistant**
University of Nebraska at Omaha, Department of Mathematics, Omaha, NE
• Developed a network model of macrophage protein interactions to study HIV infection
- 2017 – 2019 **Research Assistant**
University of Nebraska at Omaha, Department of Biology, Omaha, NE
• Used CRISPR/Cas9 to investigate the genetic basis of stress coping behaviors in zebrafish

AWARDS AND HONORS

2025	American Society of Human Genetics (ASHG 2025) – Travel Award	\$1,500
2023	Huck Institutes of the Life Sciences – Graduate Travel Award	\$2,250

RESEARCH SUPPORT

Submitted Support

R21 (MPI: Bresnahan, S. T.)

Title: *Understanding how maternal metabolic environment moderates fetal genetic regulation of the placental transcriptome underlying childhood metabolic traits*

Sponsor: National Institute of Environmental Health Sciences

Funding period: 04/01/2027 – 03/31/2029

Total costs: \$500,000

MPI: Bhattacharya, A. Huang, J.

R21 (Co-I: Bresnahan, S. T.) – under consideration for funding by NIEHS

Title: *Identifying PFAS-Induced Disruption of Placental Imprinting and Neurodevelopment Through Multi-Omic Negative Control Calibration*

Sponsor: National Institute of Environmental Health Sciences

Funding period: 04/01/2027 – 03/31/2029

Total costs: \$500,000

PI: Bhattacharya, A.

Co-I: Huang, J., Fry, R., O'Shea, M.

R01 (Co-I: Bresnahan, S. T.)

Title: *Dissecting genetic and environmental effects on the placental transcriptome in gestational complications*

Sponsor: National Institute of Child Health and Human Development

Funding period: 04/01/2027 – 03/31/2032

Total costs: \$2,000,000

MPI: Bhattacharya, A. Huang, J., Fry, R., Santos, H.

Completed Support

2019 – 2024	National Science Foundation Graduate Research Fellowship Program	\$173,000
2019 – 2022	Integrative Pollinator Ecology Graduate Training Fellowship Program	\$58,000
2019 – 2020	Penn State University Graduate Fellowship	\$19,500
2017 – 2018	UNO ORCA – Funds for Undergraduate Scholarly Experiences Grant	\$2,500

PUBLICATIONS

Publications in Peer-Reviewed Journals

1. **Bresnahan, S. T.**, Yong, H., Wu, W. H., Lopez, S., Chan, J. K. Y., White, F., Jacques, P.-É., Hivert, M.-F., Chan, S.-Y., Love, M. I., Huang, J. Y., & Bhattacharya, A. (2026). Long-read transcriptome assembly reveals vast transcriptional complexity in the placenta associated with metabolic and endocrine function. *Nature Communications*.
<https://doi.org/10.1038/s41467-026-71303-4>

Publications in Peer-Reviewed Journals (cont.)

2. Chang, Y. H., **Bresnahan, S. T.**, Head, S. T., Harrison, T., Yu, Y., Huff, C. D., Pasaniuc, B., Lindström, S., & Bhattacharya, A. Isoform-level analyses of 6 cancers uncover extensive genetic risk mechanisms undetected at the gene level. *British Journal of Cancer*. 133, 874-885. <https://doi.org/10.1038/s41416-025-03141-y>
3. **Bresnahan, S. T.**, Mahony, S., Anton, K., Harpur, B., & Grozinger, C. M. (2025). Intragenomic conflict underlies extreme phenotypic plasticity in queen-worker caste determination in honey bees (*Apis mellifera*). *Genome Biology*. 26(171). <https://doi.org/10.1186/s13059-025-03628-0>
4. **Bresnahan, S. T.**, Ma, R., Galbraith, D., Rangel, J., & Grozinger, C. M. (2023). Beyond conflict: Kinship theory of intragenomic conflict predicts individual variation in altruistic behaviour. *Molecular Ecology*. 32(21), 5823-5837. <https://doi.org/10.1111/mec.17145>
5. **Bresnahan, S. T.**, Lee, E., Clark, L., Ma, R., Rangel, J., & Grozinger, C. M. (2023). Examining parent-of-origin effects on transcription and RNA methylation in mediating aggressive behavior in honey bees (*Apis mellifera*). *BMC Genomics*. 24, 315. <https://doi.org/10.1186/s12864-023-09411-4>
6. Crone, M., Boyle, N., **Bresnahan, S. T.**, Biddinger, D., & Grozinger, C. M. (2023). More than mesolectic: Characterizing the nutritional niche of *Osmia cornifrons*. *Ecology and Evolution*. 13, e10640. <https://doi.org/10.1002/ece3.10640>
7. **Bresnahan, S. T.**, Döke, M. A., Giray, T., & Grozinger, C. M. (2021). Tissue-specific transcriptional patterns underlie seasonal phenotypes in honey bees (*Apis mellifera*). *Molecular Ecology*. 31(1), 174-184. <https://doi.org/10.1111/mec.16220>

Publications Under Review/Revision in Peer-Reviewed Journals

1. **Bresnahan, S. T.**, Xiong, C., Head, T., Chang, Y. H., Bhattacharya, A., Huang, J. Y. (2026). ICONIC: An R package for integrating instrumental variable- and negative-control-informed causal discovery and diagnostics in multiomic studies. *medRxiv* [Preprint]. <https://doi.org/10.64898/2026.08.26.26361466>
2. Head, S. T., **Bresnahan, S. T.**, Cole, N., Wu, W., Bhattacharya, B. (2025). Quantification method affects replicability of eQTL analysis, colocalization, and TWAS. *bioRxiv* [Preprint]. Under revision at *Nature Genetics*. <https://doi.org/10.1101/2025.08.20.671303>
3. **Bresnahan, S. T.**, Yong, H. E. J., Drelichman, M. G., Campbell, S. N., Trapse, A. E., Romo, G. R., Cellini, C. M., Lopez, S., Yen Chan, J. K., Chan, S. Y., Elkin, E. R., Bhattacharya, A., & Huang, J. Y. (2026). Natural variation in transplacental transfer efficiency exposes distinct transcriptional network architectures of PFAS effects on birth weight and gestational age. *bioRxiv* [Preprint]. Under revision at *Environmental Health Perspectives*. <https://doi.org/10.64898/2026.03.23.712893>
4. Head, S. T., Nemani, A., Chang, Y. H., Harrison, T. A., **Bresnahan, S. T.**, Rothstein, J. H., Sieh, W., Lindström, S., & Bhattacharya, A. (2026). Improving isoform-level eQTL and integrative genetic analyses of breast cancer risk with long-read RNA transcript assemblies. *bioRxiv* [Preprint]. Under revision at *Nature Genetics*. <https://doi.org/10.64898/2026.03.22.713514>

Non-Refereed Articles

1. Bresnahan, S. T. (2026). [*One in five observational omics papers overstates causal claims \(with caveats\)*](#). Substack.
2. **Bresnahan, S. T.** (2023). [*Metacleaner: Automated curation of barcode sequence databases for metabarcoding and metagenomics*](#). Substack
3. **Bresnahan, S. T.** (January – April 2022). [*Entomologist of the Month Factsheets*](#). Penn State Insect Biodiversity Center, College of Agricultural Sciences.
4. **Bresnahan, S. T.** (2020). [*Mind the Bees – Ralf Nauen and Colleagues Protect Pollinators Through Neonicotinoid Research*](#). Penn State College of Agricultural Sciences News.
5. **Bresnahan, S. T.** (2020). [*The “Hidden” World of Colony-Level Impacts of Neonicotinoids on Social Pollinators*](#). Penn State College of Agricultural Sciences News.

PRESENTATIONS AND POSTERS

Oral Presentations

International and National Presentations

1. **Bresnahan, S. T.**, Xiong, C., Bhattacharya, A., Huang, J. (2026, **invited**). *Negative controls and instrumental variables for identifying PFAS effects on placental gene regulation and developmental outcomes*. Society for Epidemiological Research, Phoenix, AZ.
2. **Bresnahan, S. T.**, Wu, W., Huang, J., & Bhattacharya, A. (2025). *Long-read transcriptome assembly reveals vast transcriptional complexity in the placenta associated with metabolic and endocrine function*. ASHG, Boston, MA.

International and National Presentations (cont.)

3. **Bresnahan, S. T.**, Mahony, S., Anton, K., Harpur, B., & Grozinger, C. M. (2024, **invited**). *Investigating the molecular mechanisms of intragenomic conflict in honey bees*. Biology and Genomics of Social Insects, Cold Spring Harbor Laboratory, Long Island, NY.
4. **Bresnahan, S. T.**, & Grozinger, C. M. (2024, **invited**). *Investigating the molecular mechanisms of intragenomic conflict in honey bees*. Plant and Animal Genomics Annual Meeting, San Diego, CA.
5. **Bresnahan, S. T.**, Hines, H., Zayed, A., Rangel, J., Li-Byarlay, H., & Grozinger, C. M. (2022, **invited**). *Intragenomic conflict and its epigenetic basis in honey bees*. International Union for the Study of Social Insects, San Diego, CA.
6. **Bresnahan, S. T.**, Axtell, M., & Grozinger, C. M. (2022, **invited**). *Evaluating piRNAs as a mechanism of intragenomic conflict in honey bees*. Plant and Animal Genomics Annual Meeting, San Diego, CA.
7. **Bresnahan, S. T.**, Li-Byarlay, H., Rangel, J., Ma, R., Galbraith, D., & Grozinger, C. M. (2021). *Evaluating intragenomic conflict in altruistic, pheromone-mediated honey bee behaviors*. Biology and Genomics of Social Insects, Cold Spring Harbor Laboratory, Long Island, NY.

State, Regional, and Local Presentations

1. **Bresnahan, S. T.** (2026, **invited**). *Effects of gestational PFAS exposure on placental genomic imprinting*. The University of Texas MD Anderson Cancer Center, Department of Epidemiology Trainee Works in Progress Seminar Series, Houston, TX.
2. **Bresnahan, S. T.** (2025, **invited**). *Long-read assembly and analysis of breast tissue and tumor*. The University of Texas MD Anderson Cancer Center, Department of Epidemiology Trainee Works in Progress Seminar Series, Houston, TX.
3. **Bresnahan, S. T.**, Döke, M. A., Giray, T., & Grozinger, C. M. (2021, **invited**). *Tissue-specific transcriptional patterns underlie seasonal phenotypes in honey bees*. Penn State University Center for Pollinator Research Symposium, University Park, PA.

Poster Presentations

1. **Bresnahan, S. T.**, Bhattacharya, A. (2026). *Understanding how maternal metabolic environment moderates fetal genetic regulation of the placental transcriptome underlying childhood metabolic traits*. ASHG, Montreal, CA.
2. **Bresnahan, S. T.**, Xiong, C., Huang, J., & Bhattacharya, A. (2026, **invited**). *A negative control framework for causal inference in genomic mediation: application to PFAS-induced placental imprinting and neurodevelopment*. STATGEN, Atlanta, GA.
3. **Bresnahan, S. T.**, Huang, J., Fry, R. C., & Bhattacharya, A. (2026). *Effects of PFAS exposure during pregnancy on placental gene regulation and infant neurodevelopment*. Society for Reproductive Investigation, San Juan, PR.
4. **Bresnahan, S. T.**, Wu, W., Huang, J., & Bhattacharya, A. (2025). *Long-read assembly of the placenta reduces inferential uncertainty and unveils novel isoforms associated with gestational diabetes mellitus*. Biology of Genomes, Cold Spring Harbor Laboratory, Long Island, NY.
5. **Bresnahan, S. T.**, Ma, R., Galbraith, D., Rangel, J., & Grozinger, C. M. (2023). *Kinship theory of intragenomic conflict predicts altruistic and selfish behaviors in honey bees*. International Conference on Pollinator Biology, Health, and Policy, University Park, PA.
6. **Bresnahan, S. T.**, Axtell, M., & Grozinger, C. M. (2020). *Evaluating the role of PIWI/piRNAs in intragenomic conflict in honey bees*. Regulatory and Non-Coding RNAs, Cold Spring Harbor Laboratory, Long Island, NY.

TEACHING EXPERIENCE

2024 – Present	MDACC Epidemiology Trainee Works in Progress biweekly meetings
Fall 2021	Honey Bees and Humans, Penn State, ENT 222 (TA), 90 students
Spring 2020	RNA-seq Analysis, workshop through the Penn State University Library, 20 participants
Fall 2018	Introduction to Neuroscience, UNO, NEUR 1520 (TA), 50 students

SERVICE

Ad Hoc Journal Reviewer

Nature; Heredity; Cell Genomics; Scientific Reports; BMC Genomics; Placenta; G3 (Genes|Genomes|Genetics); Genome Biology and Evolution; Molecular Ecology; Insect Molecular Biology

Student Mentorship

Mentoring responsibilities include providing training on molecular biology laboratory techniques, computational genomics and statistical methods, study design, data analysis, and presentations

Graduate Students

2025 – 2026	<u>Yung-Han Chang</u> , Doctoral Student, Department of Epidemiology, The University of Texas MD Anderson Cancer Center, UTHealth Houston Graduate School
2025 – 2026	<u>Sierra Lopez</u> , Masters Student, Global Environmental Science, The University of Hawai'i at Mānoa (mentored through Dr. Jonathan Huang's lab)
2024 – 2025	<u>Robbie Kelly</u> , Doctoral Student, Huck Institutes of the Life Sciences, PSU

Graduate Students (cont.)

2023 – 2024	<u>Avi Eliyahu</u> , Doctoral Student, Department of Entomology, The Hebrew University of Jerusalem (mentored at Penn State through the US-Israel Binational Agricultural Research and Development Fund Graduate Fellowship Program)
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Undergraduate Students

2026 – Present	<u>Charis Xiong</u> , Department of Biostatistics, Rice University
2025 – 2026	<u>Aryun Nemani</u> , Department of BioSciences, Rice University
2025	<u>William Wu</u> , Department of BioSciences, Rice University
2023 – 2024	<u>Owen Christopher</u> , Department of Biology, Penn State University
2023	<u>Mariam Taninabe</u> , Department of Biology, Penn State University

LABORATORY SKILLS

- Molecular cloning, PCR, qPCR, CRISPR/Cas9, RNAi, cell culture
- Chromatin immunoprecipitation (ChIP), assay for transposase-accessible chromatin (ATAC)
- Short- and long-read sequencing library preparation, Illumina sequencing, Oxford Nanopore sequencing

COMPUTING SKILLS

- Advanced: R (including software development), Bash, HPC environments, AWS
- Intermediate: Python, MatLab, C++

SOFTWARE

I develop, maintain, and contribute to several R packages, all available on GitHub

1. [ICONIC](#): instrumental variable- and negative-control-informed causal discovery and diagnostics for multiomics
2. [scAmbi](#): mapping ambiguity overdispersion correction and BCV analysis for scRNA-seq
3. [metacleaner](#): automated curation of sequence databases for metabarcoding and metagenomics
4. [SQANTI3](#): tool for the quality control of long read defined transcriptomes